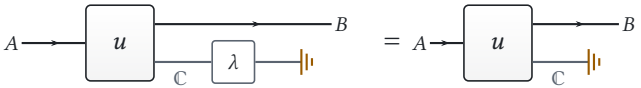
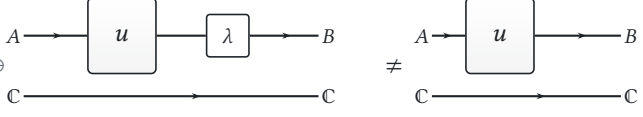
(a) a phase on the garbage is a garbage move, and the discard absorbs it



$\mathscr{E}_{\lambda u} = \mathscr{E}_u$: one garbage move,
by the unitary λ of the garbage $\mathbb{C}$

(b) beside a second summand the same phase is a relative phase



on $\frac{1}{\sqrt{2}}(|a\rangle + |*\rangle)$ the cross terms are
$\frac{\lambda}{2}|ua\rangle\langle *|$ and $\frac{1}{2}|ua\rangle\langle *|$: different channels

so $- \oplus \mathbb{C}$ is not a map on classes, and the sum of isometries does not extend to channels